Reliable Language Systems in Public Administration

A synthetic technical report for software integration testing

This is an original synthetic document created for software testing. It contains no real people, cases, measurements, or external source text.

1. Purpose and operating context

A regional service center is evaluating a language-assisted workflow for sorting incoming requests. The proposed system identifies the subject of each request, suggests a responsible team, and drafts a short internal summary. Staff members remain responsible for every decision and may revise or reject any suggestion. The evaluation therefore focuses on decision support rather than autonomous case handling.

Requests arrive through web forms, email, scanned letters, and telephone notes. They vary in length, terminology, and writing quality. Some contain several unrelated questions, while others omit information needed for routing. A reliable workflow must preserve the original message, record every transformation, and make uncertainty visible to the employee reviewing the case.

The initial pilot covers waste collection, building permits, and local transport. These domains were selected because they have distinct vocabularies and established escalation paths. The pilot does not include emergency services, legal determinations, benefit eligibility, or other decisions where an incorrect suggestion could immediately affect a person's rights or safety.

2. Data preparation and evaluation

The evaluation corpus contains synthetic requests written by the project team. Each request has a documented intended category, a short rationale, and one or more ambiguity labels. Personal names, addresses, account numbers, and health information are deliberately excluded. This design permits repeatable experiments without exposing residents or employees to privacy risks.

Performance is measured at several levels. Routing accuracy indicates whether the suggested team matches the documented category. Coverage measures how often the system makes a usable suggestion. Calibration compares confidence values with observed accuracy. Review time records whether the suggestion helps staff complete the task faster without encouraging superficial approval.

Aggregate scores can conceal important weaknesses. Results are therefore broken down by channel, topic, text length, ambiguity, and language variety. Every failed case is assigned an error type such as missing context, competing topics, uncommon terminology, or unsupported inference. The error catalogue guides changes to prompts, reference material, and escalation rules.

3. Retrieval and response grounding

Draft summaries are grounded in an approved collection of service descriptions and internal routing notes. Each retrieved passage carries a source identifier, publication date, and owning team. Material that has passed its review date is excluded until the owner confirms that it remains valid. The system displays the supporting passages beside every generated suggestion.

Retrieval quality is tested separately from generation quality. A relevant passage may be absent because the search query is weak, the index is incomplete, or the document uses different terminology. Conversely, a retrieved passage may be topically related but unsuitable for the specific request. Separate labels make these failure modes visible and prevent the generator from receiving credit for information it could not access.

The system must decline to draft a confident summary when evidence is missing or contradictory. A structured fallback message tells the reviewer which detail could not be established. This behavior is evaluated with intentionally incomplete requests and with pairs of source passages that describe different procedures for different municipalities.

4. Human review and operational safeguards

Reviewers see the original request, the proposed category, the draft summary, confidence indicators, and cited source passages in one interface. Accepting a suggestion requires an explicit action. Corrections are logged as structured feedback, but they do not enter a training set automatically. A data steward first checks whether the correction contains personal or confidential information.

Operational monitoring tracks error rates, response latency, unavailable dependencies, and the proportion of suggestions that reviewers change. Sudden shifts trigger an investigation rather than automatic adaptation. Model versions, prompt versions, index snapshots, and configuration values are recorded for each suggestion so that an incident can be reproduced.

A rollback plan keeps the previous validated configuration available. If monitoring reveals an unexpected pattern, the service can return to manual routing without losing incoming requests. Staff training emphasizes that fluent text is not evidence of correctness and that citations must be checked when a request is unusual or consequential.

5. Open questions

The pilot leaves several questions unresolved. The team has not yet established how performance changes during seasonal peaks, how well the workflow handles mixed-language messages, or how reviewers interpret confidence indicators under time pressure. It is also unclear how often source owners can maintain routing notes when procedures change rapidly.

Future work will examine accessibility, long-term reviewer behavior, and the cost of maintaining the evidence collection. A successful pilot requires more than a high accuracy score: the workflow must remain understandable, contestable, secure, and useful under ordinary operating constraints.